

Johan Naresh

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EXPERIENCE

Data Science Research Analyst

Dec 2025 - Present

(Startup) Sephira Institute - Public Sentiment Team

- Built end-to-end ETL workflows in **Python** to ingest, clean, and validate global equity and sentiment datasets, transforming raw multi-source inputs into a structured country-month panel for hypothesis-driven time-series and cross-sectional modeling.
- Performed exploratory data analysis using vectorized rolling correlations, log-differences, and OLS residualization to identify statistically robust lead-lag relationships across markets, sectors, and sentiment regimes.
- Designed and backtested a monthly rebalanced sentiment-driven investment strategy with strict no look-ahead controls, outperforming a global benchmark (CAGR **11.9% vs 9.0%**; Sharpe 0.76 vs 0.49; beta 0.73).
- Led robustness and risk diagnostics including volatility, max drawdown, and CAPM alpha in collaboration with a Cambridge professor, formalizing documentation and validation standards to ensure reproducible analytical workflows.

Quantitative Developer

Dec 2025 - Present

Wat Street - Quant Design Team

- Created a high-throughput automated **Python** pipeline to process large-scale financial news streams, leveraging **MinHash and locality-sensitive hashing (LSH)** to eliminate near-duplicate articles at scale before embedding-based clustering.
- Developed a **time-aware semantic clustering framework** that dynamically tightens similarity thresholds for temporally proximate articles, isolating true market-moving events from redundant news flow.
- Developing event-level datasets for downstream research, enabling sentiment extraction and macro signal generation from distinct financial news events rather than raw article-level noise.

Data Engineer

Apr 2025 - Mar 2026

Marble Investments

- Engineered asynchronous, semaphore-controlled API ingestion pipelines in Python (asyncio + httpx) with rate-limit handling, retry logic, and structured validation, reducing total acquisition time by **80%** across thousands of equity tickers.
- Designed concurrent batch processing and **Pandas**-based transformation workflows to standardize financial metrics across thousands of tickers, using vectorized operations to ensure consistency for downstream analytics and portfolio research.
- Built indexed **MongoDB** architectures with upsert refresh logic to enable scalable ETL processing and low-latency queries.
- Developed a **Next.js + TypeScript** internal analytics dashboard streaming live financial KPIs, enabling real-time portfolio monitoring, exploratory data analysis, and investment decision support.

PROJECTS

🔗 Real-Time Gym Rep Counter | C++, Node.js, WebSockets, Supabase

- Built low-latency rep counting in **C++** using accelerometer and gyroscope streams and defining rep-detection logic (filtering, peak/valley detection, debouncing, timing constraints), with timestamped serial ingestion via vcpkg-managed deps.
- Developed a **full-stack pipeline** where, **Node.js** backend ingests high-frequency JSON sensor streams, validates and buffers events, and sends updates via **WebSockets** to a responsive HTML/CSS/JS dashboard with real-time state synchronization.
- Connected the dashboard to **Supabase** for live session state (counts, metrics) and added **computer-vision** features.

📱 ESG Mobile App - Finalist SMAC Competition | Dart, C++, CMake, Gemini API

- Built a **Flutter** Android ESG-scoring app in a team setting with barcode/photo/text inputs; integrated the **Gemini** API to generate product scores and explanations and display them in-app.
- Implemented device persistence for user preferences/history so settings and previously scored products remain across sessions.

Career Compass Coalition - Growth & Development | Growth, Content Ops, Analytics, Static Web

- Scaled student user base from **1,000** to **17,500+** across a college-access support platform; analyzed engagement data and iterating on content, timing, and channel strategy; led growth to **24,000+** social media followers.
- Led survey-driven newsletter production and managed the editorial pipeline for **14** student writers.
- Built and deployed the organization's first centralized web resource using a modern static site architecture.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript, C++, C, HTML/CSS, Java, Dart, Bash

Frameworks & Libraries: React, Next.js, Flask, Pandas, TensorFlow, BeautifulSoup, Selenium, yfinance

Tools & Platforms: Git, GitHub Actions, Linux, PostgreSQL, MongoDB, Firebase, Vercel, Power BI, Excel, APIs (REST, LLMs)

EDUCATION

University of Waterloo

Computer Science and Finance, Bachelor of Computing and Financial Management (Honours Co-op)

- GPA:** 3.94/4.00 | *Awarded President's Scholarship of Distinction*
- Relevant Coursework:** Algorithm Design, Data Abstraction, Financial Markets & Data Analytics, Linear Algebra